

Coding Exercise: Building an open-loop motor imagery brain-computer interface

General description

This assignment consists in implementing a complete, open-loop processing pipeline of a typical brain-computer interface (BCI) based on motor imagery (MI). The final goal of the assignment is to produce a BCI model that could subsequently be used for closed-loop BCI interaction, and to evaluate this model by extracting single-trial decoding accuracy, given a pre-recorded MI BCI dataset. To accomplish this assignment you need to write MATLAB or Python code implementing suitable signal processing, pattern recognition and machine learning steps towards the construction of the BCI decoder and the classification of the given MI data for two mental tasks (right and left hand MI).

The assignment is open-ended. You have the possibility to elaborate the processing modules, perform comparisons, or go the extra mile along certain proposed or custom dimensions (i.e., examine other mental tasks beyond right and left hand MI, that are contained in the given files; produce additional useful visualizations of the data; make observations across the whole population, etc.).

Resources

Pseudonymized EEG data of 10 able-bodied subjects are provided in GDF format and can be downloaded here: [Zenodo_link_pending](#)

The biosig toolbox available [here](#). You will need the biosig functions to load GDF data in MATLAB, and you can further explore the possibilities of this open-source toolbox (as well EEGLAB Fieldtrip, MNE—in Python, and others) to implement the different processing modules of your pipeline. Exemplary MATLAB code is provided in the same [Github repository](#).

Background

Brain-computer interfaces (BCIs) are systems that enable communication with the environment and control of external devices by translating brain signals into device commands, thus circumventing the normal neuromuscular interaction pathways commonly employed by humans. Simply said, BCI realizes the “dream” of direct thought-control (brain-control) of devices (wheelchairs, robots, orthotics/prosthetics, software, smart homes, etc.). Since they are believed to be applicable even in complete paralysis, BCIs have so far mainly been envisioned as an assistive technology for people with severe motor disability, who therefore comprise BCI’s main and traditional end-user group. BCIs have been also more recently shown to play a profound role in motor neurorehabilitation. Decoding mental states in real-time can be also useful for able-bodied users (e.g., automatic fatigue recognition during driving through continuous brain activity monitoring). Electroencephalography (EEG) is the

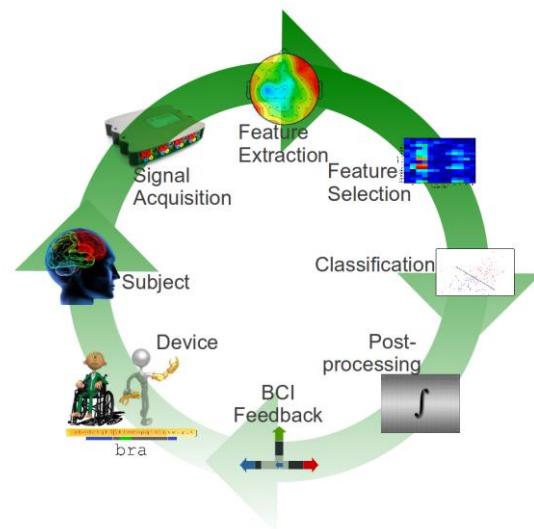
most practical way (non-invasive, safe, with excellent time resolution and adequate spatial resolution and brain coverage, minimally obtrusive, etc.) to pick up brain activity. EEG measures the electrical potential (voltage) on a certain position on a user's scalp. This measurement is generated by the electrical activity ("neuronal spikes" and "spike trains") of millions of neuronal cells within the underlying cortical networks that are in the proximity of the scalp position in question, as it propagates through the brain tissue, the skull and the scalp. For this reason, despite its great advantages, EEG is also a notoriously low signal-to-noise ratio signal; its amplitude is extremely small (in the range of 100 μ V maximum) and it is vulnerable to various noise sources (muscular artifacts, ambient electromagnetic interference, etc.).

Motor imagery BCIs are able to support self-paced (spontaneous, asynchronous and independent of any external stimuli) BCI control by decoding imagined movements (or, for motor disabled people, attempted movements) employed by a subject. MI allows BCI control thanks to the so-called Event-related Synchronization/Desynchronization (ERD/ERS) phenomenon. ERD/ERS is basically a modulation of the BCI user's sensorimotor EEG rhythms (SMRs) elicited by MI. More specifically, MI manifests either (mainly) as a decrease (ERD) or increase (ERS), with respect to an immediate previous "resting" interval, of the EEG signal's bandpower, at specific frequency bands and cortical locations. The frequency bands that are known to be associated with MI are the μ -band (approximately 8-14 Hz) and the β -band (approximately 18-24 Hz, many researchers report a broader band about 16-30 Hz). The ERD phenomenon is time-locked to the onset of MI, while ERS usually appears at the offset of MI, and is specific to the beta band (hence, also called "beta rebound").

Importantly, the cortical locations that are "activated" by MI depend on the imagined movement, consistent with the human sensorimotor cortex homunculus. The EEG SMR correlates of hand imagery are known to be mostly "contralateral" to the imagined hand side (i.e., the right brain hemisphere is activated by left hand MI, and vice-versa), although "ipsilateral" (same side) activation is also possible. Feet imagery activates mostly medial areas of the motor cortices. Overall, different MI tasks have their own distinct spatio-spectral signature, a sort of "fingerprint". The basic idea giving rise to MI BCI is to decode, in real-time, a number of MI tasks (usually, 1-4) employed by a user whose EEG are concurrently monitored, and link each decoded mental task to a specific command to an external device. For example, a user may exploit a 3-class (right and left hand, feet) MI BCI to drive a brain-actuated wheelchair; feet imagery

moves the wheelchair forward, while right/left MI accomplishes the respective turn. The wheelchair stays still when “rest”/idling is detected.

Since EEG BCI must work on “big” and multivariate raw data (e.g., 8-128 EEG channels, time resolution of up to 20 kHz--usually 256 to 1024 Hz) which may grow bigger after processing (i.e., spectral decomposition of the signals that adds an extra frequency dimension), mental task decoding can benefit a lot from state-of-the-art pattern recognition and machine learning methodology. Modern BCI systems consequently respect the “classical” pattern recognition architecture consisting of three main steps: **feature extraction, feature selection and classification**. Because EEG (and brain signals, in general) are vulnerable to noise, BCIs usually also include, at the very beginning of the processing pipeline, a signal **pre-processing** module including sub-tasks like: spectral filtering (e.g., to remove powerline noise with a notch filter, to limit the investigation to the bands of interest with a band-pass filter, or to simply remove the DC component of certain segments of interest), spatial filtering (to remove common noise affecting all channels with a Common Average Reference filter, or to help the specific channel activity to emerge removing the influence of neighboring areas with a Surface Laplacian filter). The figure above illustrates the main processing modules of the BCI loop (note that pre-processing is included in signal acquisition).



Tasks

You need to build an MI BCI processing pipeline given the aforementioned MI EEG data. Exemplary MATLAB code for possible processing operations is provided on [Github](#).

You need to write MATLAB code (or Python, but examples are only provided in MATLAB) in the form of a main script `processMI.m` (equivalently, `processMI.py` in Python) which implements all the necessary processing modules and produces suitable visualizations (plots, tables, bar graphs, etc.). You can implement your solution either as a single script that embeds the entirety of your code in a single function/script or (ideally) pursue a structured programming approach creating separate auxiliary functions for each distinct processing operation or module.

Your code should implement all of the following processes and modules:

1. Trial extraction.

2. Pre-processing operations.

This includes any set of suitable artifact removal, spatial, spectral and/or temporal filtering methods that you know are useful for MI BCI.

3. Feature extraction.

4. Feature selection/dimensionality reduction.

5. Classification and post-processing.

Apart from the implementation of a suitable classification model (i.e., training/parameter estimation and application to get classification accuracy), you should make sure to employ a sound training and testing methodology (i.e., cross-validation, training/testing set splitting).

Your code should further produce the following types of results:

6. Evidence of the quality and sanity of the recorded EEG signal for one or more subjects.
7. Grand average spectra per mental task for one or more subjects. Reflect whether any differences can be observed and, if so, whether these are found in the anticipated bands and locations.
8. Topographical maps of feature fitness for salient frequency bands. Reflect on whether any emerging brain patterns are located on the anticipated brain regions given the mental tasks employed by subjects.
9. Concentrated classification accuracy results per subject. Based on this, reflect on the potential of a prospective BCI user to get in control of a 2-class, EEG-based MI BCI.

The final classification results must be presented individually for each subject. Thus, it is advised that extracted trials of the same subject which are acquired from different GDF files are concatenated in a single data structure that is fed to subsequent processing modules. Importantly, accuracy must be presented at the single trial level. You are free (and encouraged) to process the data in sliding windows (within each trial). This approach would allow for the real-time, closed-loop BCI to work at greater time resolution (i.e., “refresh”/decision rate) than the N s that each trial lasts in the given dataset. However, in this case you must use some sort of post-processing of your single-sample/window classification decisions to also come up with a final prediction for each trial, so that accuracy can be reported at the trial level. You may want to choose decoding models that embed feature extraction and/or selection into the classifier, such as deep neural networks.

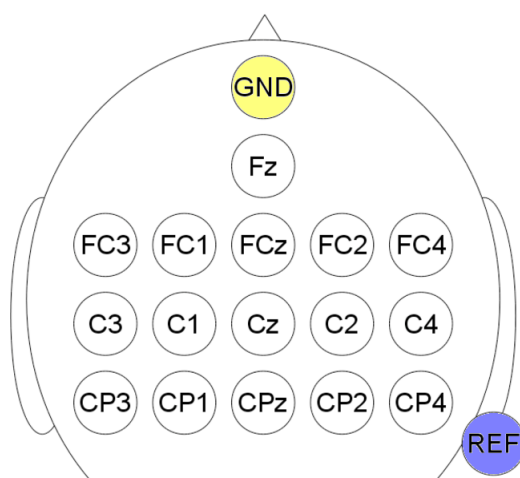
You may want to extend your investigation along any of many possible dimensions. Possible topics for additional investigation may include:

- The shared data also contain feet MI (in certain cases, also a resting/idling class). You can examine different 2-class MI pairs (in terms of both classification outcome and visualization), like Right Hand vs Feet and Left Hand vs Feet. Check if you can find subjects who performed poorly with Right vs Left Hand, but perform better with another combination.

- Consider producing 3-class and 4-class (where possible) classification accuracy (enabling 3 or 4 commands).
- Examine the effect of the number of used features on accuracy and over-fitting. You can explicitly check for overfitting by comparing training and testing set accuracy.
- Produce full confusion matrices instead of just the final classification accuracy and comment on biasing effects.
- Compare different filtering, feature extraction, feature selection and classification methods with respect to their effects on the final accuracy.
- Try training a classifier on 9 subjects and testing on the 10th. Does this kind of model training (subject-unspecific) work? Compare the classification accuracy of this simple transfer learning approach with the one acquired with the subject-specific classifier.
- Using the fixation cross intervals as “resting”/idling segments (or, where available, also the actual resting trials), make an estimation of the false positive rate of the MI BCI system you have implemented, thus estimating your system’s capacity of implementing intentional non-control.

Information on the data and the experimental protocol

Since the location of an EEG signal stream on the scalp (given its proximity to a certain cortical area) is of vital importance in decoding mental tasks, the international 10-20



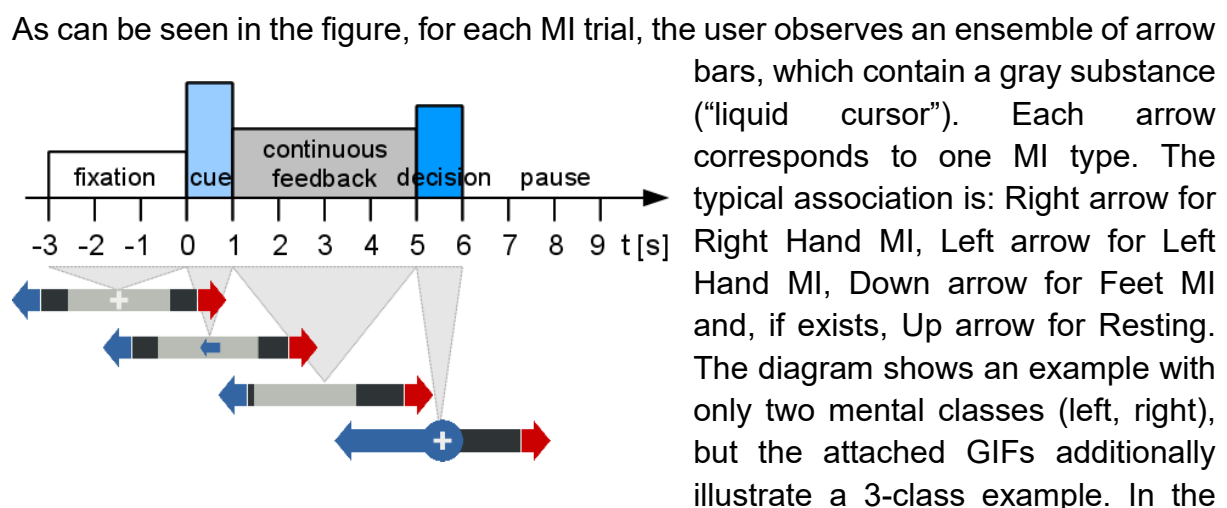
positioning system is used to standardize the placement of EEG channels on commercial EEG caps. The EEG data provided for this assignment contain 16 channels placed as shown in the picture. The order of channels in the data array of the GDF files provided, when loaded in MATLAB, derives from this electrode/channel map through row-wise numbering (i.e., Fz is the first channel, C3 is the 7th, and CP4 is the 16th channel). The data have been recorded with a gTec gUSBamp amplifier (gTec, Austria) at 512 Hz sampling rate (i.e., the potential at each channel is

measured and stored 512 times within each second) and a hardware bandpass filter within [0.1 100] Hz (i.e., EEG signals were recorded after being already filtered in this band at the amplifier level).

The data provided come from a classical “offline” (open-loop), “bar feedback” MI protocol. Offline/open-loop means that EEG signals of several repetitions of different types of MI are collected, but without real-time processing of the signal or any real feedback to the user regarding decoding. This kind of protocols are conventionally used to collect EEG data and “labels” (i.e., annotations of what mental task each segment of EEG represents), so as to build a BCI model that can be later used for

actual BCI control (i.e., closed-loop and “online”, with real-time EEG processing and decoding, where subjects are actually in control of the feedback and/or some brain-actuated device).

In this protocol, users perform several (usually, 3-4 in an experimental session) “runs”. A run is a block of continuous EEG collection corresponding to a single saved GDF file. Each of these runs contains 15 trials for each of the contained MI tasks. The files provided contain right hand MI, left hand MI, feet MI, and some of the files also contain “resting” trials (i.e., trials where the user is idling, not performing any MI task). The naming convention of the GDF files provided is SubjectID.Date.Time.offline.mi.mi_X.gdf. In this database, X is either **rlbf** (or **rlf**), for files containing right hand, left hand and feet MI, or **rlsf**, for files additionally containing resting trials. Each new MI trial follows the previous one and is visualized through a bar feedback the trained user is looking at on a computer screen. The following diagram provides the timeline of a trial with the visualization signaling used at each event.



beginning of a new trial, the user observes a “fixation cross”, which informs the subject they should focus, stop producing artifacts (speaking, moving, blinking) and anticipate the upcoming MI trial. After 2-3 sec, the fixation cross is replaced by a colourful arrow (called the “cue”) that informs the user which MI type he should perform. After 1 s, the liquid cursor starts moving towards the direction of the respective arrow, until, after 4 seconds it reaches the threshold (end or the corresponding arrow bar). This visualization provides “positive” fake feedback to prepare the user for the closed-loop, online runs, where the liquid cursor only moves according to the decoding output of the BCI model (i.e., the user is in brain-control of the cursor, the “liquid” moves towards the correct direction only if the BCI’s decoding is correct).

EEG data in GDF format can be loaded in MATLAB (or Python) with the biosig toolbox. For MATLAB, after adding biosig to the path, the data can be loaded with the command: `[data, header] = sload(filepathname);` EEG data are then loaded in the workspace in the format (time samples x EEG channels).

The information on the timing of the protocol's events that point to the type of task performed by the user at each EEG segment is saved in the file's header. Specifically, these markers/triggers can be found in the struct variable `header.EVENT`. In some of the files, this information may also be present in an additional 17th channel (trigger channel). `header.EVENT` includes 3 variables: `TYP`, `POS` and `DUR`, which determine the type, the position (i.e., "when", at which time samples the event in question happened) and the duration (in time samples) of an event, respectively. The events of interest stored correspond to salient events of the aforementioned visual protocol. Here is the glossary of the events included in the provided data:

1: Trial start (does not correspond to some visual event)

786: Fixation cross appearance

769/770/771/783: (arrow) for Left Hand, Right Hand, Both Feet, Rest, respectively

781: Liquid cursor starts moving

Of note, no "trial end" event is marked, because the duration of the liquid cursor movement is always exactly 4 seconds (this info can be also retrieved from the `DUR` variable).